Université Paris Sorbonne Nord
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Master 1.
Distributed Java Programming
TP4: Sockets

This exercise uses the class `Teacher` defined in TP2.

1. Define a class `Server` featuring the following fields and methods:
   - an attribute `pool` of type `ExecutorService`;
   - attributes `port` and `poolSize`;
   - a boolean attribute `isFinished`, as the server will run until a specific request is received (say "`shutdown`");
   - a constructor `Server(int port, int poolSize)` that creates the server socket and launches the executor service;
   - getters for all the fields; and
   - a method `manageRequest()` that accepts and processes requests with the aid of the executor service and the class `Slave` below.
2. Define a class `Slave` that implements the interface `Runnable` for managing the requests. This class must feature:
   - an attribute `socket` for the connection with the client;
   - a constructor with one parameter to initialize the socket to the client; and
   - a method `run()` that will read the request sent by the client (e.g. name/identifier of the operation and its parameters). Then, depending on the operation, calls the appropriate methods in `Teacher` and sends back the result to the client.

Define two projects, one for the server and one for the client:

1. For the server, define a class `Main` with only one method `main` that initializes a set of teachers and creates an object of type `Server` with pool size of 10.
2. For the client, define a class `Main` that tests your implementation by sending some requests to the server.

The service offered by the server should provide operations for:

- Create new teachers.
- List all the teachers of a given discipline.
- Computing the mean of salaries of the teachers of a given discipline.
- Return the number of teachers of a given discipline.
- Shutdown the server (and no more request can be accepted).